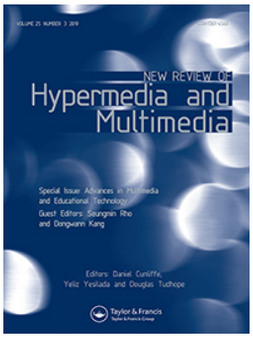

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Systematic review and usability evaluation of writing mobile apps for children

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ABSTRACT
Writing apps for children have been developed in recent year to support effective learning and writing skills of students. In this work, we perform a systematic review of all currently available apps for the operating system iOS and Android. We consider a number of newly release children writing apps by focusing on their features like the range of functions, target user groups, operating language, acquisition costs, user ratings and the connection between acquisition cost and user ratings. In total, we analysed 497 apps based on functions offered and their cost to users. In addition to the systematic review of apps, we also perform a participant-based usability evaluation of a representative sample i.e. 8% sample of the total collected apps. Totally 40 apps are selected for usability evaluation including 20 apps for iOS and the remaining 20 apps for Android. As a result of usability evaluation of apps, it is found out that with the increase in the number of functions in a mobile app, the usability of the app decreases. The subjective evaluation reveals that participants felt better user experience with iPhone in comparison to Android enabled devices.

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Systematic review; usability evaluation; writing mobile apps; apps for children.

1. Introduction

In this age of digital mobile technology, almost everyone owns a smart phone either he/she can afford it or not. According to Fennell, Barkley, and Lepp (2019), cell phone ownership among adults has exceeded by 90% of its numbers in population. Technology made a very big change in education world with the likelihood of creating more educational opportunities. Both

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stakeholders i.e. teachers and students are positively affected by this technology occupation. Teachers are searching the ways to use technology in their class work and students showing more interest while using technology in learning (Wu, Li, & Liu, 2019). The growing number of mobile apps (see Figure 1) has given rise to the phenomenon of mobile learning i.e. m-learning.

In today's context, m-learning is considered the latest type of learning introduced as a result of this technological revolution, in which new learning options are provided through mobile applications (Ebrahim, Ezzadeen, & AK, 2015; Leinonen, Keune, Veermans, & Toikkanen, 2016). With the assistance of a wireless and ubiquitous system of learning, smartphone users can use the educational material on their devices to learn. Such educational materials can be in the form of multimedia content or learning objects, which are more interactive, highly visual, engaging and effective for learner quality of learning (QoL) (Yousafzai, Chang, Gani, & Noor, 2016). Indeed, these learning objects can be easily used by the students whenever necessary and wherever they are by maximising this "always there, always on" technology (Ally, 2009). Ideally, smart devices accompanying applications (apps) can create exciting and effective learning environments for learning and instruction in early childhood (Goodwin, 2012; Papadakis, Kalogiannakis, & Zaranis, 2017). Developers are increasingly creating educational apps that target this age group (Hiniker et al., 2015). A content analysis in the "Education Section" of Apple app store, conducted by Shuler (2009) found that almost half (47%) of the top 100 selling apps targeted preschool and primary school age children, and mathematics and literacy apps were the most popular categories of apps. In a study conducted later (Shuler, Levine, & Ree, 2012) found that the number of apps which targeted preschool and primary school age children increased to 72% from 47% in 2009. In Android and iOS Operating Systems both, educational apps are among the top three categories of apps most accessed by the users in 2018 (ThinkMobiles, 2019).

Along with several leaning apps available online, leaning writing apps are one of the most beneficial apps for making children learn writing. For children in early days of their learning writing, writing apps provide an excellent platform for learning writing by tracing or drawing alphabets and numbers in a fully interactive environment. This kind of learning method is very exciting for children who can actually get bored in the conventional style of learning. Writing apps can be used in schools in supervision of their teachers and also used in homes for preschool learning under supervision of parents. Teachers and parents can also use these apps to get idea about the process of writing and similar practices can be performed on paper for children. Children learn through tracing or writing pattern of letters and numbers writing in a multimedia environment with phonic sounds of each alphabet and number and other music. Mostly there are also other functions available augmenting the process of learning writing like recording function wherein children can record and listen voices to make practice.

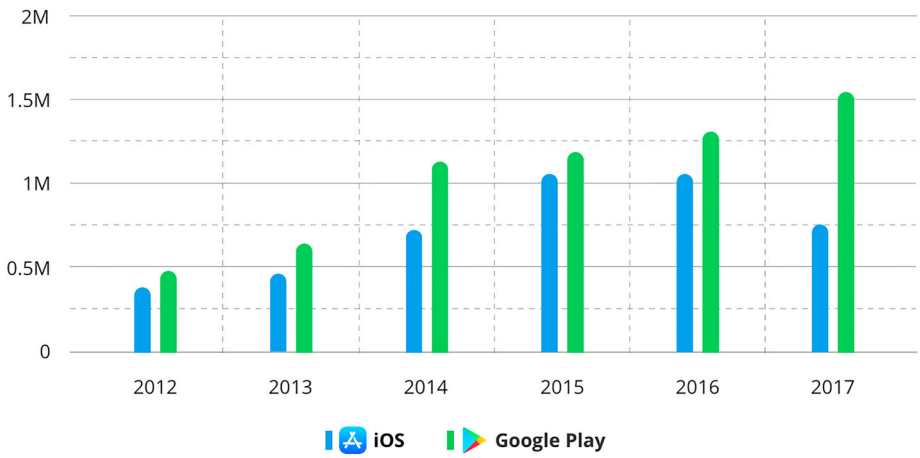


Figure 1. Annual number of apps for iOS versus Android platforms.

In this paper, our main focus is to do systematically review all available apps related to children learning writing on both operating systems i.e. iOS and Android. The review is conducted for writing apps available between June 2018 and December 2018. Besides this, we also perform a detailed usability evaluation of a sample of mobile apps. The purpose is to give research and developers community an insight into a very important type of apps which can help improving the m-learning experience of children with writing apps. With so much work being done on mobile application development, there is a need for such type of work identifying the usability problems in mobile apps. According to the best of our knowledge, many researchers have worked on a review of mobile apps but no work exists on this category of apps. The rest of the paper is organised as follows: Section 2 provides related works in this area including other systematic reviews of mobile apps. Section 3 details our review search and filtering criteria and discusses the characteristics of the apps reviewed. In section 4 we provide the usability evaluation setup and criteria of comparing various writing apps. Section 5 provides the results of our analysis in terms of comparisons, merits, drawbacks of different features of these apps. Finally, Section 6 concludes this review.

2. Related works

In this section, we discuss the works that are related to our work. It has been found out that there is not much work existing for writing mobile apps. Therefore, we highlight some similar attempts done for other set of applications along with few works relating to writing apps.

The only work found relating to writing apps was done by Kanala, Nousiainen, and Kankaanranta (2013) who explored the use of the prototype of a mobile application for the enhancement of children's motivation for writing under a

project funded by Finnish Funding Agency of Technology and Innovation (Tekes). The results are explored from students' and experts' perspectives. The findings suggest that the use of a mobile application can have the potential in supporting children's creative writing skills and their motivation to complete writing tasks.

2.1. Usability models for mobile apps for children

In the rest of the section, we discuss major usability work done on mobile applications. We start with the work of Masood and Thigambaram (2015), which investigates the importance of usability in user interface (UI) design of mobile educational applications (MEA) designed for children aged 4–5 years. Usability testing was conducted to with four children to evaluate the UI design based on children's mental model in addition to the quality of their learning experience. An eye tracking glasses was utilised to capture the children's eye movements while the usability heuristics were used to collect the descriptive data regarding the interface design. Results indicated that usability study can be considered as a multi-stage problem solving process where it analyse and forces how the children use a product and actual first time users' intuition of their design experience and learning curve. Kraleva, Kralev, and Kostadinova (2016) also focused on the possibilities of using mobile learning in the Bulgarian preschool education for young children. Two conceptual models of applications for mobile learning at preschool children are designed.

Dyslexia is one of the most common learning disabilities experienced by children and adults. Skiada, Soroniati, Gardeli, and Zissis (2014) present the methodology, environment setup, design choices, implementation and the results of preliminary evaluation and assessment of "EasyLexia", a mobile application for children with learning difficulties. Preliminary results show the promising prospects mobile learning holds in such contexts. The main focus of their work is to develop a mobile application which could foster learning and help children improve some of their fundamental skills, such as reading comprehension, orthographic coding, short-term memory and mathematical problem solving. We attempted to design a stimulating and interactive experience for children, which could encourage the learning process.

Chau (2014) show that for children's mobile apps to be developmentally meaningful, they need to satisfy three conditions:

- Apps must be designed appropriately to accommodate the developmental stages and needs of young children.
- Content must be designed to promote young children's development in the areas of cognition, academic skills, social-emotional skills and physical development.

- Digital interaction engages children in activities and behaviours that foster optimal development assets.

Chau (2014) devised three instruments to evaluate and examine the breadth, depth and design quality of 100 children's apps from the apple store for iPad. Content analysis revealed that only a non-significant majority of apps (58%) were meaningfully designed for preschool children in terms of user interface, audio and visual design and instructional support. The apps selected for this sample included games and learning activities, interactive eBooks, as well as creativity and utility apps. The contents of these apps tended to cluster around school skills and they rarely engaged children in activities beyond academic drill and practice. These apps largely ignored the social, emotional and physical aspects of children's development. Using numerous vignettes and examples as illustrations, the analysis highlights design techniques, content offering and technological featured that could be productive toward children's development, as well as those that distract from meaningful user experience. This study underscores the need for developmentally meaningful children's mobile apps for preschool children.

2.2. Related systematic reviews of mobile apps

In Tan et al. (2017), the authors utilised a scoping process to identify and analyses 49 crises informatics articles that focus on mobile apps in disaster situations. The study investigates the various mobile apps that engage with the crowd during disaster situations. Findings show that apps used in disasters can be general-purpose apps or built-for disaster purpose apps. The review further focuses on the built for disaster-purpose apps and shows the various interactions these apps foster with the public.

Grist, Porter, and Stallard (2017) systematically appraised the available research evidence on the efficacy and acceptability of mobile apps for mental health in children and adolescents younger than 18 years. The method of systematic searching for relevant publications between January 2008 and July 2016 in the following different databases was undertaken: APA Psych Net, ACM Digital Library, Cochrane Library, Community Care Inform-Children, EMBASE, Google Scholar, PubMed, Scopus, Social policy and practices, web of science, journal of medical internet research, Cyber psychology, behaviour and social networking, and OpenGrey.

In Holmen, Wahl, Småstuen, and Ribu (2017), the authors systematically review studies that aimed to evaluate integrate communication with mobile apps for tailored feedback between patients with diabetics and health care personnel in terms of:

- Study characteristics
- Functions

- Study outcomes
- Effects
- Methodological Quality

A literature research was conducted following the international perspective registers of systematic reviews protocol, searching for apps with integrated communication for persons with diabetics tested in a controlled trial in a period 2008 to 2016.

There are several studies pertaining to the usability of mobile commerce (m-commerce) applications, however, these studies do not sufficiently address issues about usability methods or adequately provide knowledge concerning usability methods used in most of the empirical usability evaluation for applications in m-commerce. In Hussain, Mkpojiogu, and Kamal (2016), a systematic review on usability evaluation methods for m-commerce apps attempts to review available literature with the aim of capturing the usability techniques commonly or frequently used in the domain of usability evaluation for applications in m-commerce. To achieve the stated research goal the study applied systematic literature review methodology. Sixty seven (67) papers in the area of usability evaluation for m-commerce apps were downloaded. Out of these papers, 21 most relevant studies were selected for a review in order to extract the appropriate information needed for the analysis.

A systematic review of effects on learning and development by Herodotou (2018) identified 19 studies reporting learning effects on children 2 to 5 years old. The number of children participating in experimental, quasi-experimental, or mixed-method studies was 862 and in descriptive or correlation studies, 941. The majority of studies reported positive effects on literacy development, mathematics, science, problem-solving and self-efficacy. Among the factors explaining observed effects were design features, the role of adults and a similarity between applications and transfer context. Although drawing firm conclusions remains a challenge, this review forms the first step towards systematic research in the field and contributes to shaping directions for future research.

Harrison, Flood, and Duce (2013) argue that attributes such as cognitive load, tend to be overlooked in the mobile usability models that are most prominent despite their likely impact on the success or failure of an application. Therefore, they introduced the PACMAD (People At The Centre Of Mobile Application Development) usability model which was designed to address the limitations of existing usability models when applied to mobile devices. PACMAD brings together significant attributes from different usability models in order to create a more comprehensive model.

3. Systematic review of mobile applications

In this section, we describe the systematic review of writing apps for children available for both Android and iOS. The systematic review is discussed in different sub-sections below.

3.1. Application searching and filtering

We search for relevant apps on Android and iOS using pertinent keywords in English. We used the following Keywords for this purpose: writing apps, learning writing for kids, preschool learning apps, writing for kids, improve learning, toddler learns to write, basic learning, how to write, writing instructions, writing motivation, calligraphy, scribble. Each app is manually checked for relevancy to writing apps. There is no audit policy on Google store currently while on apple store, apps are first internally reviewed by an app review board before their deployment.

Apple app store provides better search facility for application search in comparison to Google play store. Search results on Apple app store can additionally be sorted by relevance, popularity, user ratings and the date of release. We searched for writing apps for children in Apple app store using keywords as mentioned above during November 2017 to December 2017. An app is selected as far it is relevant ignoring the fact that if it is available only for iPhone or iPad or for both. Additionally, we checked if the app is only available for Android operating system or an adapted version is also available for iOS. The total relevant hits for search on Apple app store are 249 apps. Every app is reviewed in detail according to App name, developer information, language, last update, developer information, availability on devices and availability on desktop application, acquisition cost, popularity/user ratings, interfaces and range of functions, user groups and number of functions per apps. Similarly, we search for relevant writing mobile apps on Google play store using keywords as mentioned above. Google app store does not provide a facility to filter search results for apps according to individual needs. Furthermore, the given "number of hits" is not only the number of apps but also the number of detected search terms in the app title and the app description. Thus the search term, "writing apps" is led to hits that are not relevant mostly. This search for writing apps on Google play store was done during July 2017 to September 2017. We collect the same information for Android apps as for iOS apps. Additionally, every app was crosschecked for the availability of an iOS version. Altogether, we found 246 app relevant to writing apps available for Android operating systems.

We found total 497 apps during the review process. After a detailed relevance search filtering, a set of relevant available apps available exclusively for the operating system iOS are 251 while apps available exclusively for the operating system Android are 246 and apps available on both operating systems are 56.

3.2. General criteria for systematic review

Table 1 shows general criteria to be used for systematic review of writing apps (Arnhold, Quade, & Kirch, 2014).

3.3. Annual development of app release

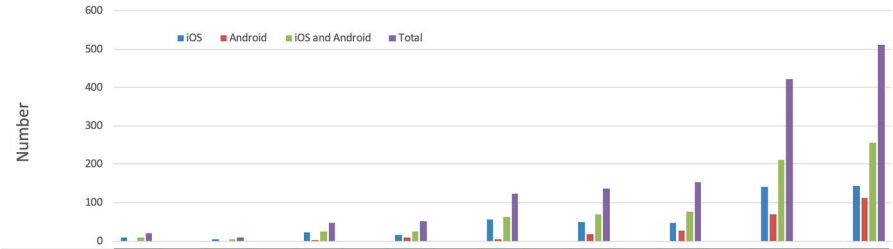
The development of first writing apps for children on Apple app store dates back to year 2009 while for Google play store, it is year 2011. With the passage of time, number of developed writing mobile apps for children increased annually with 10 mobile apps in year 2009 on Apple app store and 142 apps in year 2017. Not surprisingly, the number of Android apps was more rapidly increasing on Android than iOS as shown in Figure 2. In Apple app store, some apps were specifically designed for iPhone and some designed for iPad and some of them were designed for the both iPhone and iPad. It can be observed in Figure 2, the number of apps are increasing progressively as the years pass.

3.4. Operating languages

Table 2 summarises the availability of apps according to operating language. Majority of the apps available for iOS are in English (97%, 245/251) while 89 % of the apps exclusively running on Google play Store (89%, 235/246) are in English. Apps using other languages were relatively lower in number for both operating systems. For iOS, only 3 % (3%, 06/251) of the apps using other language than English while for Android Operating system, this number is a bit larger i.e. 11 % (11%, 11/246).

Table 1. Table to test captions and labels.

Category	Sub-category
General Information	Application Name
	Application Language
	Date of release or latest update (the acquisition of the release date was only possible for iOS apps; for Android apps, only the date of the latest update could be recorded)
	Availability of a desktop application
Operating System	App exclusively for the iOS operating system
	App exclusively for the Android operating system
	App for both operating systems available
Developer Information	321
Acquisition Cost	Freeware
	Exact price
	Availability as "lite" version (paid apps sometimes offer free or cheaper lite versions with limited functionality)
Popularity/User Ratings	Number of downloads/installations
	User rating
	Number of user ratings
Range of functions (Multiple Selection Possible)	Description/Information Function
	Write/Trace alphabets and numbers function
	Customised setting function
	Sharing and exporting function
	Reward function
	Phonic Sound Function
	Learning Through Games
	Record
Target User Groups	Children/Students
	Parents/Teachers
	Both user groups



	2009	2010	2011	2012	2013	2014	2015	2016	2017
iOS	10	5	22	17	57	50	48	142	144
Android	0	0	2	9	5	19	28	69	112
iOS and Android	10	5	24	26	62	69	76	211	256
Total	20	10	48	52	124	138	152	422	512

Figure 2. Annual development by operating systems.

3.5. Acquisition cost

The majority of iOS apps are available as a lite version (a smaller scale version of an app) while full scale versions of the apps are to be purchased. Apps available on Google play store are mostly free and also available as a full version and very rare apps are required to be purchased. Table 3 describes the details of free and paid writing apps on both operating systems.

The analysis of apps price distribution in Table 3 reveals that the quantity of free apps available is much greater than of paid apps. The number of free apps available for both operating systems i.e. iOS and Android make up to 94 % of the total apps (93.6%, 484/517). The percentage of free apps available on android is 97.1% (239/246) with paid apps making up to 2.8 percent (07/246). Similarly, the number of free apps available for iOS make 97.6 % (245/251) with the proportion of paid apps reaching to 2.3 % (06/251).

In Table 4, proportions for free apps are given year by year for both operating systems. It can be observed that in duration 2011-2103, all writing apps available on both operating systems were free. The proportion of paid apps contributed in later years.

3.6. Range of functions

The analysis of the number of functions available in learning writing apps is given in Table 5. It can be observed that a large number of writing apps (32.3% – (161/497) are limited to contain one function only. Almost 27 % of the apps (27.3% –

Table 2. Apps distribution by operating language on different operating systems.

	iOS	Android	iOS and Android	All
Total Apps	251	246	56	553
English	245	235	52	532
Others	6	11	4	21

Table 3. Price distribution of writing apps by operating systems.

	iOS	Android	All
Free	245	239	484
Paid	6	7	13
Paid Liter Versions	21	6	27

136/497) apps combined two functions while only 16 % of the total apps offered three functions (16% – 80/497). Applications containing four or more than four functions make almost 12 % of the total apps (12.4% – 62/497). Overall, it has been observed that android based applications offer more functions as compared to iOS based apps.

3.7. Function descriptions

Figure 3 describes the range of functions available in writing apps. It can be observed that the Description/information function is available in 89.5% (266/497) of the apps. This function describes that an app contains some kind of detailed description about its usage, its features and price, its availability as a lite versions and other technical details. Another function "the Write/Trace alphabets and Numbers" is offered in 333 out of 497 apps (i.e. in 67% apps). The "Write/Trace alphabet and Numbers" function tells about the facility of writing/tracing of alphabets e.g. A, B, C, etc. and numbers 1, 2, 3 etc. Another function i.e. "Customization Function" is offered in writing apps where a user can change its settings according to their own way, i.e. change in colours of pencils, background or any other changing related to sound, mode of change, rating pattern, generation of report or not etc. Almost 5.3 percent of the apps provide this function. Some apps provide a detailed customised setting and some do not provide it at all. Another function i.e. "Sharing and exporting function" is available in 6.8 % apps (34/497). This Sharing and Exporting Function helps user to share the app or its results on online social media such as the Facebook, twitter, Gmail and WhatsApp etc. Another function offered by writing apps is "reward function" (5.6% – 28/497). The objective of the this function is to reward the user after completion of the successful performance of any activity or task, excellent writing, the same pattern match, etc. It motivates the children to do better in their further tasks of the app. Figure 4 show representative examples of these major functions.

Table 4. Development of free apps since 2011.

Year	iOS	Android	All
2011	22	2	24
2012	17	9	26
2013	57	5	62
2014	47	18	65
2015	48	26	74
2016	139	69	208
2017	114	108	222

Table 5. Function distribution by operating systems.

Year	iOS	Android	All
1 Function	91	70	161
2 Functions	59	77	136
3 Functions	28	52	80
4 Functions	18	19	37
More than 4 Functions	9	6	15

3.8. Target user groups

Learning writing apps target Two types of user groups i.e. children and teachers or parents. Table 6 describes the number detail according to target user groups. Most of the writing apps are designed especially for the children while a smaller portion of apps are designed specially for parents or teachers.

4. Usability evaluation of writing apps

To analyse the usability of currently available writing apps for kids, we used the user based usability evaluation (Barnum, 2010; Bruun, 2007). Usability evaluation takes time, especially when conducting user-based laboratory think-aloud evaluations as described by Rubin and Chisnell (2008). According to their guidelines, the following six stages have to be completed when conducting such a test:

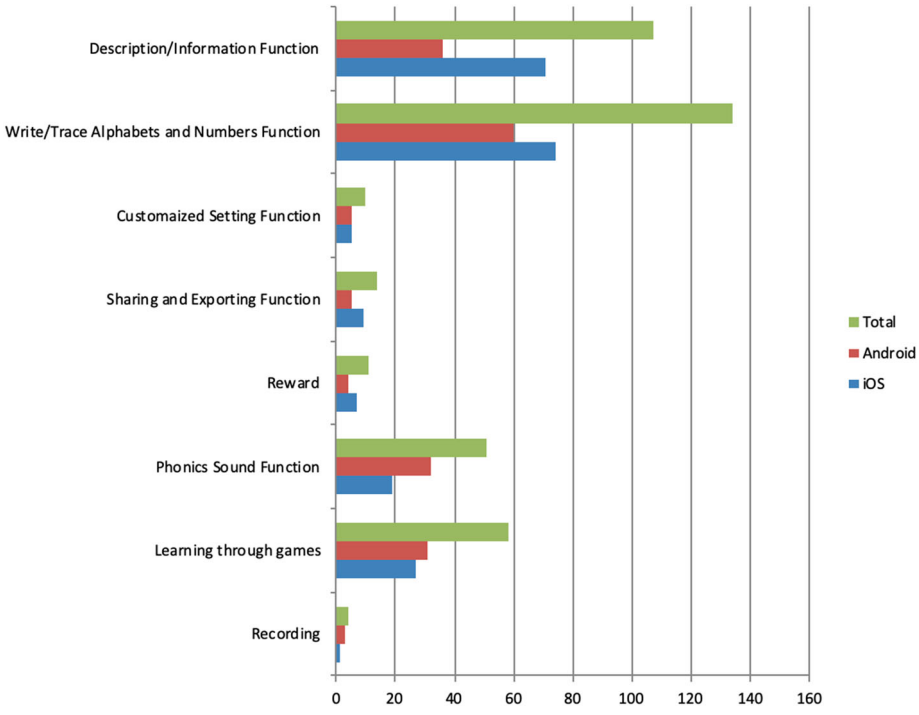


Figure 3. Range of functions by operating systems.

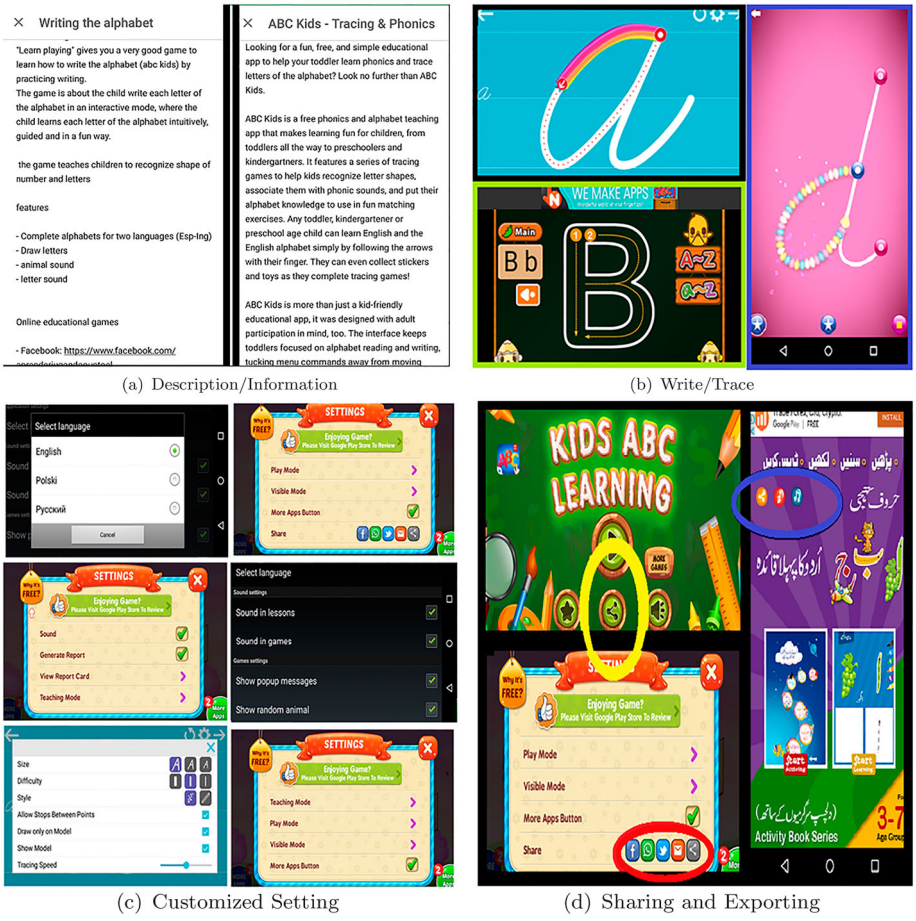


Figure 4. Representative examples of writing apps for children and their range of functions. Screenshots of: (a) Description/Information, (b) Write/Trace alphabets and numbers, (c) Customised setting, (d) Sharing and exporting functions.

- Develop the test plan,
- Select and acquire test participants,
- Prepare the test materials,
- Conduct the test,
- Debrief the participants,
- Transform data into findings and recommendations.

4.1. Participant selection

The selection of participants for usability testing is a very important phase. It is true that children are the main target audience of learning writing apps but parents and teacher also make a significant user group for writing apps. Therefore, keeping teachers and parents as part of the participant team would be a rational decision. Hence, our selection of test participants consists of children, teachers and

Table 6. Distribution of apps by target user groups.

Target User	iOS	Android	Number of Apps
Students	201	214	415
Parents/Teachers	14	17	31
Students and parents/Teachers	30	14	44

parents. All selected participants are supposed to have some experience of a period of more than 6 months or more with the usage of smart phones. Desired age for children participants is fixed at 4–6 years while the age of 30–45 years is fixed for parents and teachers category of participants. Total 12 participants are selected with 06 from children category while 03 from parents and 03 teachers category each. The reason for selecting this number of participants is based on the precedence taken from previous work. For instance in Dalkıran, Aker, Öztemiz, Taşkın, and Tunç (2014), total of 5 people were used to perform usability testing. Similarly, in Masood and Musman (2015), 8 people were chosen to perform the usability analysis. According to usability testing handbook (Goto, 1999), 4 to 8 participants are enough to perform usability testing. In our case, we chose larger number of participants because of the varying nature of target audience.

4.2. Application selection

Due to higher number of apps availability, we choose a sample of 08 % apps (total 40 apps with twenty for Android and twenty for iOS) making sure that the selected sample includes apps for both operating systems i.e. iOS and android. The sample is chosen randomly with the only surety that sample includes all apps giving all range of functions provided by all apps. The precedence of choosing 8 to 10 % of apps for usability testing exists in the literature (Arnhold et al., 2014) which justifies our decision of testing with 08 % of mobile apps. Table 7 lists all the apps selected for usability evaluation.

4.3. Test material

A questionnaire is designed to be filled by participants during or after the usability test of each selected app. Each questionnaire is filled for each single app by each participant and hence 40 questionnaires are filled by each single user. The questionnaire is prepared in accordance with the evaluation criteria defined for each app as mentioned in Table 8. Due to large number of apps to be tested, we plan the process of usability testing to be completed in several days to avoid mental stress. The questionnaire for children are filled by live observations during the tests and from recordings of the tests. All participants are asked to use think-aloud technique (Nørgaard & Hornbæk, 2006) during usability evaluation.

Table 7. Selected list of apps for usability testing.

Android Apps
Kids Write ABC!- Free Game for kids and family
Handrite Note Notepad Lite
Cursive Writing Wizard Letters
ABC kids-Tracing & phonics
LetterSchool-Learn To write the ABC Alphabet
Write Numbers-Tracing 123
ABC 123 Read write practice
Prescool Learning Games kids
Learn To Write Letters Numbers
Write Numbers-Tracing 123
Alphabet board
Learn English A to Z Activity
Mathematics and Numbers Kids
Learning Urdu for Kids
Learning To Write. Calligraphy
123 Numbers -- Count & Tracing
Kids Writing-Educational Game
Alphabet writing for kids
Writing ABC-Learn Alphabets
Kids Cursive Writing Capital
iOS Apps
Learn to write ABC-Handwriting for pres
Handwriting Worksheets for children
Preschool and Toddler kids learning Abby
Cursive Writing Capital letters free
ABC Writing Wizard Books-Kids learning
123 Kids GAMES: Math and Alphabet Games
Tracing ABC Writing Literacy
My First Alphabet Phonics ABC
Alphabet Tracing-Endless Learning
Kids Phonics A-Z, Alphabet, Letter
ABC Writing Practice Lite
ABC Bingo Song for Kids: learn alphabets
ABC Alphabet Phonics- Preschool games
ABC My First Alphabet WorkBook
ABC Typing Learning Writing Dotted Al
Sight Words:Kids Learn
ABC Kids-Tracing and Phonics
abc PocketPhonics Lite
Kids Can Spell- Writing First Words
ABC Games for Kids-EduKittyABC

4.4. Usability testing

The criteria for usability evaluation includes features like interaction process, educational and design features of content, etc as given in Table 8. Participants are given questionnaires and they are expected to provide their answers on Likert scale or on dichotomous scale (YES, NO). The scores are calculated from the mean of the respective sub criteria in each categories considered.

Table 9 shows average rating of the apps as received by usability test participants with respect to different sub-criterion on both operating systems individually and overall (marked as "All"). Each criterion has some specific set of questions in the questionnaire. For example, the values in the table reveals that usability participants are divided on questions related to Educational criterion. While there is a likelihood of more positive voting on iOS apps, participants have also voted for

Table 8. Usability assessment criteria for writing apps for children.

Main criteria	Description	Assessment criteria
Educational Features	App facilitates children to write capital and small alphabets	5-point Likert Scale
	App provides writing counting (i.e. 1, 2, 3) as well as pronounce facility	5-point Likert Scale
	App facilitate learning by playing games.	5-point Likert Scale
	App improves learning and reading skills	5-point Likert Scale
	App improves Writing Skills	5-point Likert Scale
	App contains enough activities with variety	5-point Likert Scale
Design Features	Uses sounds or music where appropriate	5-point Likert Scale
	App contains bright colours that attract children.	5-point Likert Scale
	Use of images and animation match with children skills	Dichotomous Scale (Yes, No)
	Provides proper menu for touch screen	5-point Likert Scale
Ease of Use	App is easy to use	5-point Likert Scale
	Icons sizes are appropriate and easy to handle.	5-point Likert Scale
	Provides appropriate text size, spacing etc	5-point Likert Scale
	Children know if they make a mistake	Dichotomous Scale (Yes, No)
	App crash, hang or freeze	5-point Likert Scale
	After adults support, children can use it independently	5-point Likert Scale
	It is easy to get in or out of any activity at any point	5-point Likert Scale
Comprehension	The use of appropriate Language	5-point Likert Scale
	The use of appropriate content i.e. familiar concepts	5-point Likert Scale
Pedagogic Feedback	App provides appropriate feedback	5-point Likert Scale
	App is responsive to any input	5-point Likert Scale
Navigation and Orientation	App provides clear and consistent navigation	5-point Likert Scale
	App provides easy scrolling for navigation	5-point Likert Scale
Customization	App allows for Customization and Personalization	5-point Likert Scale
Gestures Interaction	App provides appropriate gesture interaction	5-point Likert Scale

Education related usability issues on Android apps. But this confusion or division among usability participants remains limited to Educational criterion only. It can be observed that overall users have preferred usability of iOS writing apps on Android writing apps for rest of the sub-criteria.

5. Analysis of results

5.1. Comparative analysis of main criteria with respect to usability

First of all, we want to see how main criteria are contributing towards the overall usability of an application. In Figure 5, it can be observed that subjective usability evaluation score range lies between 2.9 and 4.4 in general. The highest score is obtained for "Educational criterion" while the lowest score is obtained on "Pedagogic feedback". This graph also shows that how developers are more focused on functional requirements of their product while neglecting the usability of main aspects of the product. With only one aspect of writing apps crossing rating 4 is itself an evidence of lack of focus on usability aspects of the apps.

5.2. Usability with respect to operating systems

The usability evaluation has been conducted on the two most popular operating systems (i.e. iOS ad Android) to see how each facilitates usability of the apps. iOS

Table 9. Usability scores assigned by participants (averaged).

Criteria	Android	iOS	All
Educational	4.40	4.40	4.40
App facilitates children to write capital and small alphabets	4.00	4.60	4.30
App provides writing counting (i.e. 1, 2, 3) as well as pronounce facility	3.9	4.00	3.95
App facilitate learning by playing games	4.8	4.40	4.60
App improves learning and reading skills	4.5	5.00	4.75
App improves Writing Skills	4.90	4.30	4.60
App contains enough activities with variety	4.80	4.10	4.45
Design Features	3.8	4.00	3.91
Uses sounds or music where appropriate	3.80	3.90	3.85
App contains bright colours that attract children.	4.00	4.10	4.05
Use of images and animation match with children skills	3.70	4.10	3.90
Provides proper menu for touch screen	3.50	4.20	3.85
Ease of Use	3.40	4.40	3.95
App is easy to use	3.80	4.00	3.90
Icons sizes are appropriate and easy to handle	3.00	4.70	3.85
Provides appropriate text size, spacing etc	3.10	4.80	3.95
Children know if they make a mistake	3.20	4.50	3.85
App crash, hang or freeze	3.80	4.60	4.20
After adults support, children can use it independently	3.80	4.40	4.10
It is easy to get in or out of any activity at any point	3.70	3.90	3.80
Comprehension	3.00	3.40	3.20
The use of appropriate Language	3.10	3.20	3.15
The use of appropriate content i.e. familiar concepts	2.90	3.60	3.25
Pedagogic Feedback	2.50	3.30	2.90
App provides appropriate feedback	2.60	3.10	2.85
App is responsive to any input	2.40	3.50	2.95
Navigation and Orientation	3.80	3.20	3.50
App provides clear and consistent navigation	4.00	3.00	3.50
App provides easy scrolling for navigation	3.60	3.40	3.50
Customization	3.60	4.20	3.90
App allows for Customization and Personalization	3.60	4.20	3.90
Gesture Interaction	2.50	3.50	3.00
App provides appropriate gesture interaction	2.50	3.50	3.00

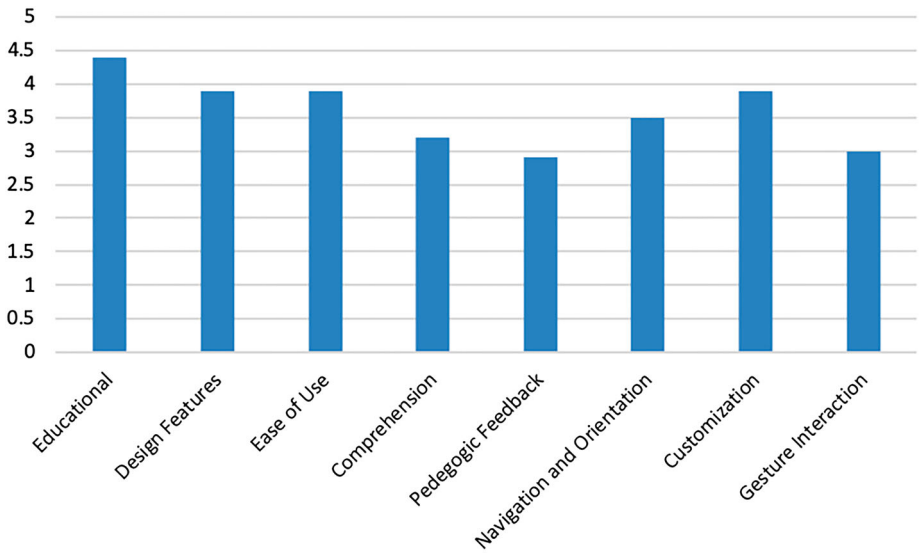


Figure 5. Usability according to main criteria.

and Android enabled devices are used for this purpose. Both devices match in their screen size, RAM and main memory, processing power to avoid any bias. Figure 6 is a line graph between ratings obtained for apps on both operating systems. Each point denotes the ratings achieved on concerned sub-criterion listed on *x*-axis. It can be seen that we get better results for iOS (red line) as compared to Android (blue line). Interestingly, there is a visible rise for android device on the usability of "Navigation and Orientation" criterion which possibly can be justified. We know that Android being open source is more pervasive in the smart phone market and this is the reason Android enabled devices are less expensive. Hence, it has more diffusion in the consumer market and probably people are more aware of the navigation scheme of Android enabled devices. This might be the possible reason behind this higher usability rating for this particular criterion.

5.3. Usability decreases as number of functions increase

Comparing usability and number of functions can be an interesting analysis. In Table 10, we have computed correlation coefficients to analyse the relation between number of functions and different usability criteria as designed for our experiments. Results show that an increase in the number of functions in an app negatively affect the overall usability of the app especially doing more damage to usability of educational features, design features and gesture interaction. Customization and pedagogic feedback also negatively correlate with number of functions in an app but these are the least damaged with increase in number of functions. However, it has been observed that increase in number of functions positively affects the usability of comprehension features with a correlation coefficient of 1. This is also shown in Figure 7

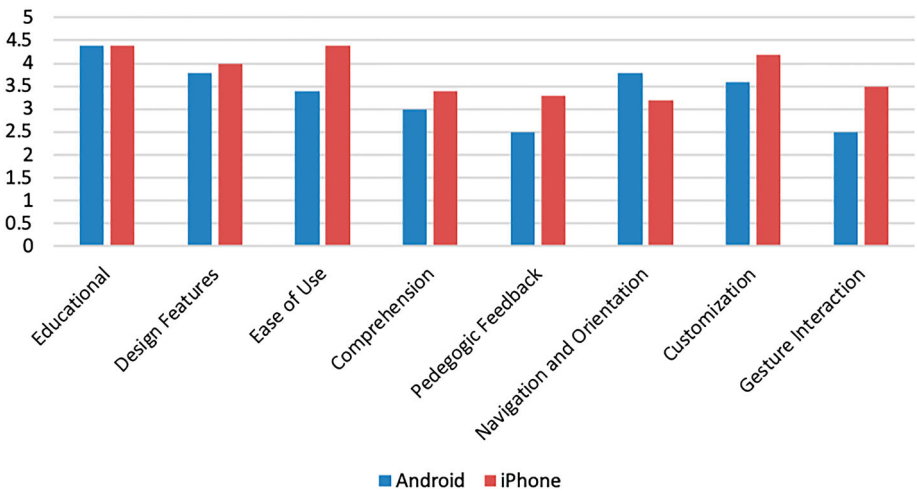


Figure 6. Usability comparison with respect to the criteria for iPhone versus Android.

Table 10. Correlation coefficients comparing number of functions with main usability criteria scores.

# Functions	Main Usability Criteria Scores							
	Educational	Design Features	Ease of Use	Comprehension	Pedagogic Feedback	Navigation/Orientation	Customization	Gesture Interaction
_	−.98	−.92	−.67	1	−.16	−.89	−.18	−.94

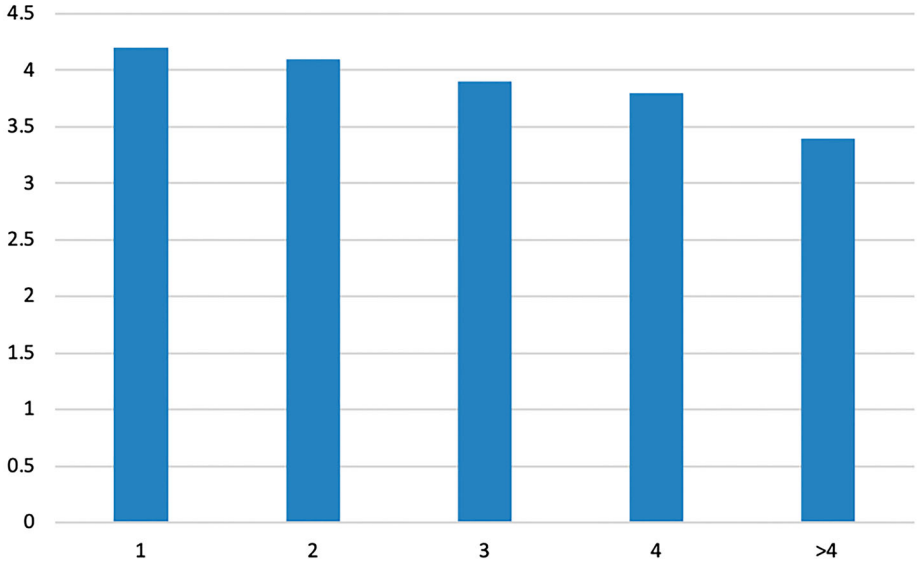


Figure 7. Usability decreases as number of functions increase.

where usability can be seen decreasing (y-axis) as number of function increase (x-axis).

5.4. Usability with respect to user type

Experimenting with several types of users provide us the opportunity of evaluating usability according to user type. In Figure 8, it can be seen that teachers perform better on all types of usability criteria. This comes from the fact that teacher have dual experience of usage of smart phone apps in general as well as usage of educational apps especially learning writing apps. Hence, experience gets an edge in this user type analysis as it can be seen in the case of parents as well. Parents rate the apps less usable than teachers but their performance is still commendable as it shows a little bit of experience of using such mobile apps which is the evidence of their interest in their children studies. Ironically, the main audience of these apps i.e. children reports more usability problems in all criteria which definitely show the degree of improvements needed on usability issues while designing educational apps.

5.5. The best vs. the worst rated app

We compare ratings of selected apps to conclude which app has been rated the best. Ratings of Android apps shown in Figure 9 while for iOS apps is shown in Figure 10.

For iOS best rated app is “Handwriting Worksheets for children” and worst rated app are “My First Alphabet Phonics ABC” and “Sight Words:Kids Learn”

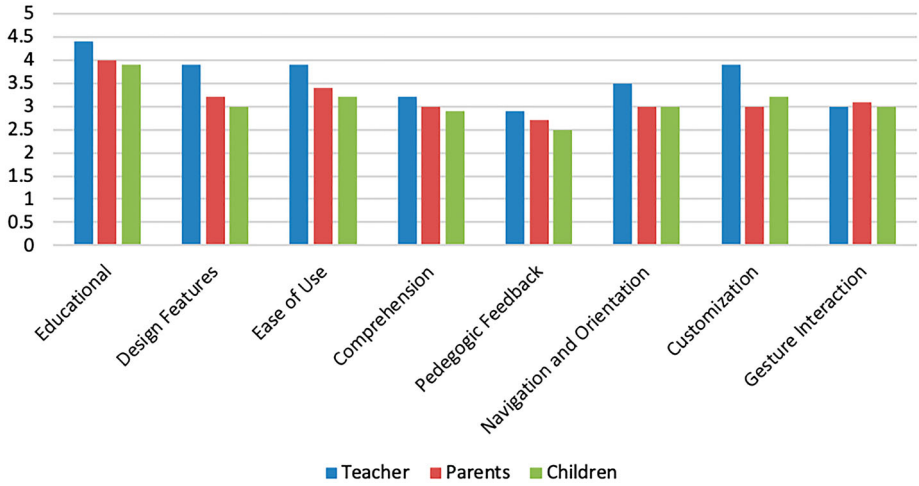


Figure 8. Overall usability with respect to the type of users.

(both got 3.2 on usability scale). Similarly for Android, best rated app is "Kids Cursive Writing Capital" and worst rated app is "Mathematics and Numbers Kids".

It is interesting to see that ratings for iOS apps are quite stable among different apps while significant amount of variations exist for ratings on Android apps. This shows how Apple store is strict in implementing the set standards for iOS application development. However, overall ratings for both operating systems remains between 3 and 4 almost.

5.6. Usability recommendations

We recommend the following usability guidelines for development of learning writing apps as a conclusion of usability evaluation.

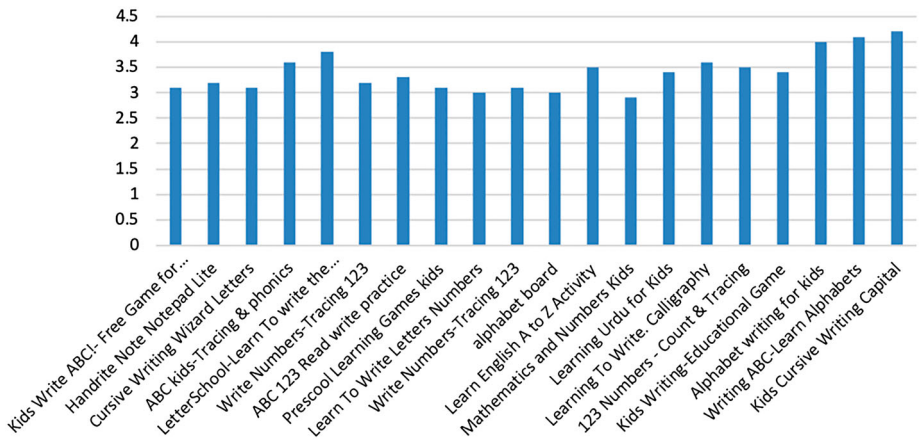


Figure 9. Usability ratings for Android apps.

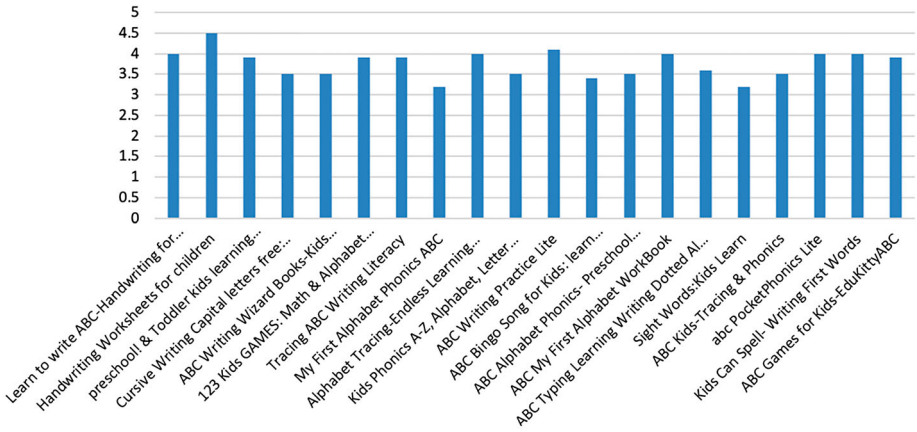


Figure 10. Usability ratings for iOS apps.

- Learning writing apps fail to provide feedback for inputs most of the time. Major reason for this could be use of lot of multimedia stuff in design of the apps. Keeping multimedia helps children understand better so it is not advised to remove multimedia content. However, what can be recommended is to keep application light weighted with respect to functions provided. Lesser the functions in an app, a better usable interface with proper feedback.
- The second most ignored point specifically in learning writing apps and generally in all children specific apps is the use of interaction gestures. Children having less cognitive capabilities cannot guess the use of gesture unless they are appropriately labelled. For example, it is always good to provide an "lens" icon with plus and minus sign for zoom in and zoom out gesture.
- Another major problem identified in apps is the use appropriate language that children can understand. Most of the developers make use of same set of icons or language that is used for any other application ignoring the comprehension skills of children. If someone is not decisive about the use of a particular term, using an iconic picture can be is recommended.
- Another major problem identified is the scrolling and navigation in learning writing mobile apps. In some apps, the tactile surface of the screen has not been used but a physical scroll bar has been given to scroll down which actually reduces the surface area of scrolling surface hence causing difficulties for children.

6. Conclusions

In this paper, we conducted a detailed systematic review and usability evaluation of learning writing mobile apps for both Android and iOS operating systems. The systematic review involves the detailed review of the operating languages used in these apps, price analysis and user type analysis etc. We further

performed a participant-based usability evaluation of a sample of apps. Participants are hired for usability testing of a selected set of apps to be tested under some defined criteria using subjective metrics. This paper shared some interesting results of the usability testing. For example, we found out that as we increase number of functions in app, its usability tends to decrease. It is also found out that participants felt that iOS apps are more user friendly than those developed for Android operating system. Usability evaluation also finds out that learning writing apps still require a lot of improvements to increase user experience for children (the major audience of these apps). It was observed that teachers and parents had a better user experience than children. Even all the participants were supposed to perform same set of usability tasks during test sessions but it was a bit tough experience for children. This paper lists out some major usability recommendations to improve user experience of children in learning writing apps to be developed in future.

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